

Question	Answer	Marks																
2(a)	One mark per mark point MP1 Correct mantissa MP2 Correct exponent Mantissa <table><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr></table> Exponent <table><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>	0	1	1	0	1	0	1	1	1	0	1	1	1	0	1	0	2
0	1	1	0	1	0	1	1	1	0									
1	1	1	0	1	0													
2(b)	One mark per mark point for working (Max 2) - number converted to binary e.g., positive binary version of 25.3125 = (0)11001.0101 - two's complement version bits flipped and 1 added = 100110.1011 -32 + 4 + 2 + 0.5 + 0.125 + 0.0625 // -32 + 4 + 2 + 1/2 + 1/8 + 1/16 - movement of binary point seen (5 places) One mark per mark point - correct mantissa - correct exponent Mantissa <table><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td></tr></table> Exponent <table><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr></table>	1	0	0	1	1	0	1	0	1	1	0	0	0	1	0	1	4
1	0	0	1	1	0	1	0	1	1									
0	0	0	1	0	1													